Sensorimotor Rhythm or Aggregate Power? An Interpretability Audit of EEGPT on Motor Imagery

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We audit what EEGPT, an EEG foundation model, encodes about motor imagery. Its pretraining pairs masked reconstruction with a representation alignment objective meant to capture consistent, high signal to noise structure. A linear probe on its frozen embeddings decodes the task about as well as feature engineered baselines. Despite that design, the embedding geometry is dominated by aggregate, largely aperiodic signal power. A parameterized, spatially controlled analysis leaves only a small, consistent trend toward the sensorimotor mu rhythm, not a strong rhythm specific representation.

The EEG power spectrum separates into two components that are both clinically meaningful, but at different granularities. The periodic component, the oscillations such as the sensorimotor mu and beta rhythms, indexes specific functional processes [7]. The aperiodic 1/f component is coarser: its slope tracks the overall balance of excitation and inhibition, a biomarker in its own right, but a broadband one rather than a marker of any particular process [5]. Recent audits report that EEG foundation models lean on the aperiodic background, and the broadband power that comes with it, more than on the specific oscillations that carry task meaning [1, 2, 3, 4]. A model that reaches accuracy through aggregate power can still be useful, but it is not grounded in the specific rhythm that clinical and BCI interpretation of motor imagery ideally rests on [7]. We extend this line of audits to EEGPT (NeurIPS 2024), a model absent from them and architecturally distinct: it adds a spatio temporal representation alignment objective to masked reconstruction [6], making it a useful test of whether that architectural change alters what the model encodes. Answering it carefully matters, because raw band power conflates periodic and aperiodic activity [9] and spatial averaging can inflate apparent rhythm specificity, so the conclusion depends on doing the spectral decomposition and the spatial control properly.

We audit the frozen EEGPT encoder on PhysioNet motor imagery, imagery versus rest, across 35 subjects (2891 cued four-second trials, ~83 per subject, each a single imagery or rest period), using the 512 dimensional embeddings as they are used downstream. We decode imagery versus rest, and we run representational similarity analysis (RSA) [8] comparing the geometry of the embedding space to per channel spectral references. To avoid the band power conflation, we parameterize each trial spectrum into aperiodic (1/f) and periodic components [5, 10] and build the oscillatory reference from aperiodic adjusted power per channel, for motor bands (mu, beta) and non motor control bands (theta, gamma). We correlate against the embedding geometry controlling for task and for the aperiodic component, contrast central against occipital channels on mu, whose central versus posterior scalp distribution separates the sensorimotor rhythm from posterior alpha [7], and test the mu effect against a trial shuffling null.

A linear classifier on the frozen embeddings decodes imagery versus rest at $\approx$ 71% within subject (chance 50%), on par with feature engineered baselines, and transfers modestly across subjects (leave-one-subject-out $\approx$ 60%, chance 50%, 31 of 35 subjects above chance), within the modest range reported for cross-subject motor imagery [11] and consistent with a representation dominated by aggregate power. Its representational geometry is dominated by aggregate power: it correlates with total signal power at Spearman $r \approx 0.3$, higher than with any single band. Once the aperiodic component is removed and the spatial construction is held fixed, only a small and consistent trend toward the sensorimotor mu rhythm remains. Aperiodic adjusted mu power is tracked $\approx$ 0.04 above the non motor control bands (mu above theta $p \approx 0.002$, mu above gamma $p \approx 0.0003$, in roughly 80% of subjects), it is stronger over central than occipital channels and survives controlling for occipital mu ($p < 0.001$); this central scalp distribution is consistent with sensorimotor mu rather than posterior alpha [7], and it exceeds a trial shuffling null ($p \approx 0.001$). Beta shows no reliable effect. We read the mu result as a trend rather than an established component: at $n = 35$ it is detectable but small, and whether it holds at larger samples is open, since significance reflects detectability, not magnitude.

EEGPT, despite its added alignment objective, represents motor imagery mainly through aggregate, largely aperiodic power, with only a small mu trend on top. This extends the aperiodic and aggregate power bias reported for reconstruction based models to a hybrid architecture, and it carries a methodological caution for the audit literature itself: the evidence for rhythm specificity depends on the analysis, because raw band power conflates periodic and aperiodic activity [9] and spatial averaging inflates apparent specificity. The balanced reading is that EEGPT reliably captures a coarse scale signal, aggregate and largely aperiodic power, which is itself physiologically meaningful rather than noise [5], but it has not yet shown strong specificity to the finer sensorimotor rhythm, despite an architecture meant to capture consistent structure. This coarse but not fine pattern may reflect current self supervised pretraining broadly rather than EEGPT specifically. EEGPT pairs masked reconstruction with a JEPA-like latent alignment, yet the coarse grounding persists. That points beyond any single objective: the same rhythm versus power lens should be applied to autoregressive, GPT-style models [12] and to joint-embedding predictive (JEPA) models [13] alike. Whether the small mu trend strengthens with more subjects, and whether such objectives can be steered toward finer oscillatory structure, are open questions we are pursuing.

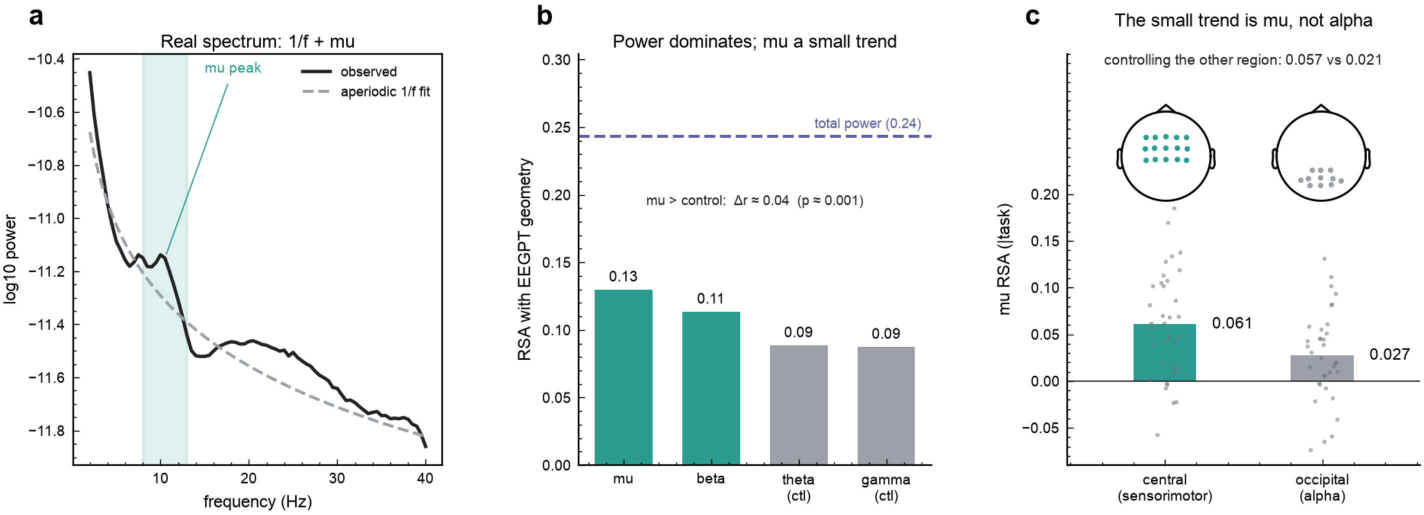

Figure 1. Auditing EEGPT on motor imagery (PhysioNet, n = 35 subjects, 2891 cued 4 s trials). (a) Grand-average sensorimotor spectrum with its fitted aperiodic 1/f component and the mu peak above it. (b) The embedding geometry tracks aggregate total power (dashed, r ≈ 0.24) far more than any aperiodic-adjusted band; mu sits only ≈ 0.04 above the non-motor controls. (c) Mu and posterior alpha share the 8-13 Hz band, so the trend is tested against location: alignment is much stronger over central than occipital electrodes and survives partialling out the other region (dots are subjects). Electrode regions are channel space, not source localization.

References

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